

Total No. of Questions : 6]

SEAT No. :

P5007

[Total No. of Pages : 2

B.E. /Insem. - 14
REFRIGERATION AND AIRCONDITIONING
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *All questions carry equal marks.*
- 2) *Use of non programable scientific calculator is allowed.*
- 3) *Assume suitable data if necessary.*

Q1) a) With schematic diagram explain the working of evaporative cooler. Compare evaporative cooler with air conditioner. **[6]**

- b) A heat pump is used to keep a room at 25°C by rejecting heat to an environment at 5°C. The total heat loss from the room to the environment is estimated to be 45,000 kJ/h and the power input to the compressor is 4.5 kW.

Determine

- i) the rate of heat absorbed from the environment in kJ/h.
- ii) the COP of the heat pump.
- iii) the maximum rate of heat supply to the room for the given power input, and
- iv) the second-law efficiency of the cycle. **[4]**

OR

Q2) a) Explain the air conditioning plant for multiplex with respect to the following's: **[6]**

- i) refrigerant
- ii) refrigeration system components
- iii) air distribution
- iv) design conditions

P.T.O.

- b) A 5TR refrigerating machine based on Bell coleman cycle operates between 5 bar and 20 bar. The air temperature after the heat rejection to the surrounding is 37°C and air temperature at compressor inlet is 7°C. The isentropic efficiencies of compressor and expander are 48% respectively. Determine [4]

- i) COP
- ii) Power per ton of refrigeration
- iii) Mass circulation of refrigerant in kg/h
- iv) Carnot COP for same air entry temperature to compressor and expander. [6]

Assume $C_p = 1.0 \text{ kJ/kg K}$, $C_v = 0.718 \text{ kJ/kg K}$.

- Q3)** a) How the refrigerants are classified? Explain the application of any one of the antifreezing solutions. [4]

- b) A domestic refrigerator used R134-a refrigerant and runs on VCC to keep inside temperature at - 10°C by rejecting heat to an environment at 50°C. The heat leakage from the environment is estimated to be 1200kJ/h and the vapor is super-heated before compression by 4°C outside the evaporator. [6]

Determine

- i) the rate of heat rejection in condenser in kJ/h,
- ii) the COP of the refrigerator.
- iii) power input to compressor and
- iv) volume displacement of compressor

OR

- Q4)** a) Describe briefly ODP and selection of environment friendly refrigerant [4]

- b) A 2TR air conditioner based on simple saturated vapor compression cycle operates between 12°C and 50°C. Determine [6]

- i) COP
- ii) Power per ton of refrigeration
- iii) Mass circulation of refrigerant in kg/h
- iv) Dryness fraction after throttling

Q5) a) Mention with the help of p-h charts various combination of evaporators and compressors in vapor compression system. [4]

b) Draw and explain in detail actual vapour absorption cycle. [6]

OR

Q6) a) Mention the advantages of Cascade refrigeration system over two stage vapour compression refrigeration system. [4]

b) A data pertaining to refrigeration system is as follows: [6]

Two separate evaporators E_1 (5 TR) and E_2 (10TR) at temperatures 5°C and 15°C respectively. Common condenser is at a temperature 38°C . The liquid from the condenser is saturated. After evaporator E_2 , pressure reducing valve bring the pressure to the suction pressure of the compressor. Single compressor and individual expansion valves are used in the system. Use Refrigerant data provided. Make the neat flow diagram and a cycle diagram on P-h plane, Find;

i) Power required to run the system

ii) COP of the System

Refrigerant Data Q6								
T	P	v_f	v_g	h_f	h_g	h_{fo}	s_f	s_g
$^\circ\text{C}$	bar	m^3/kg		kJ/kg			kJ/kgK	
5	3.63	0.000725	0.0475	40.69	189.65	148.96	0.1586	0.6943
15	4.91	0.000743	0.0355	50.10	193.79	143.69	0.1915	0.6902
38	9.14	0.000794	0.0191	72.56	202.51	129.95	0.2654	0.6830

R134-a Data								
T	P	v_g	h_f	h_g	s_f	s_g	$C_{p\text{liq}}$	$C_{p\text{vap}}$
$^\circ\text{C}$	bar	m^3/kg	kJ/kg	kJ/kg	kJ/kgK	kJ/kgK	kJ/kgK	kJ/kgK
-10	2.0052	0.09963	186.78	--	0.9509	1.7337	--	0.842
-6	2.3418	0.08591	192.03	395.15	0.9707	1.7310	--	0.858
2	3.145	0.0647	202.68	399.84	--	1.7263	1.341	0.892
12	4.4289	0.04636	--	405.51	1.0579	1.7215	1.374	0.939
24	6.4566	0.03189	233.05	411.93	--	1.7169	--	1.006
30	7.7008	0.02667	241.65	414.94	1.1432	1.7149	--	1.044
50	13.177	--	271.59	423.63	1.2373	1.7078	1.569	1.218

